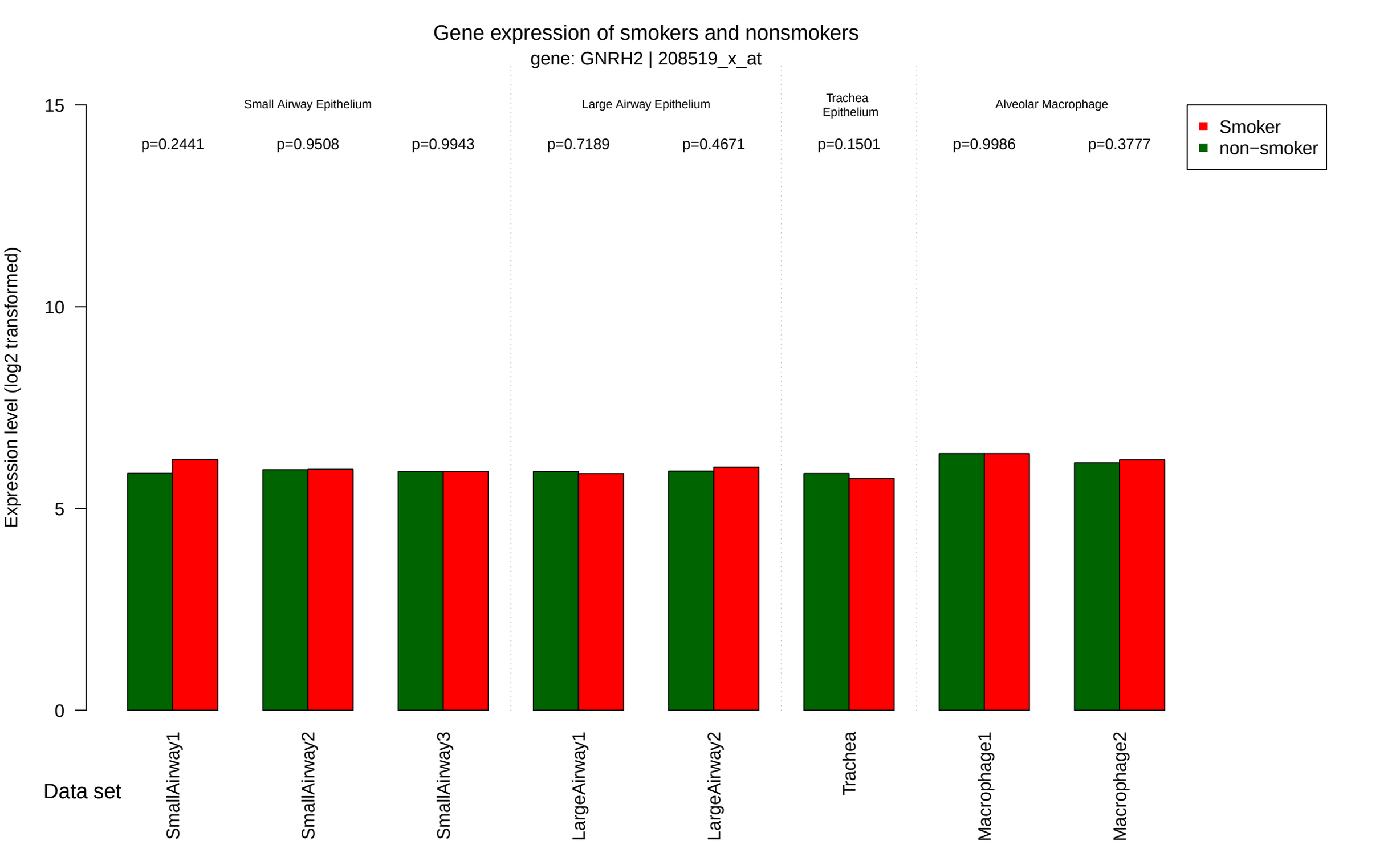
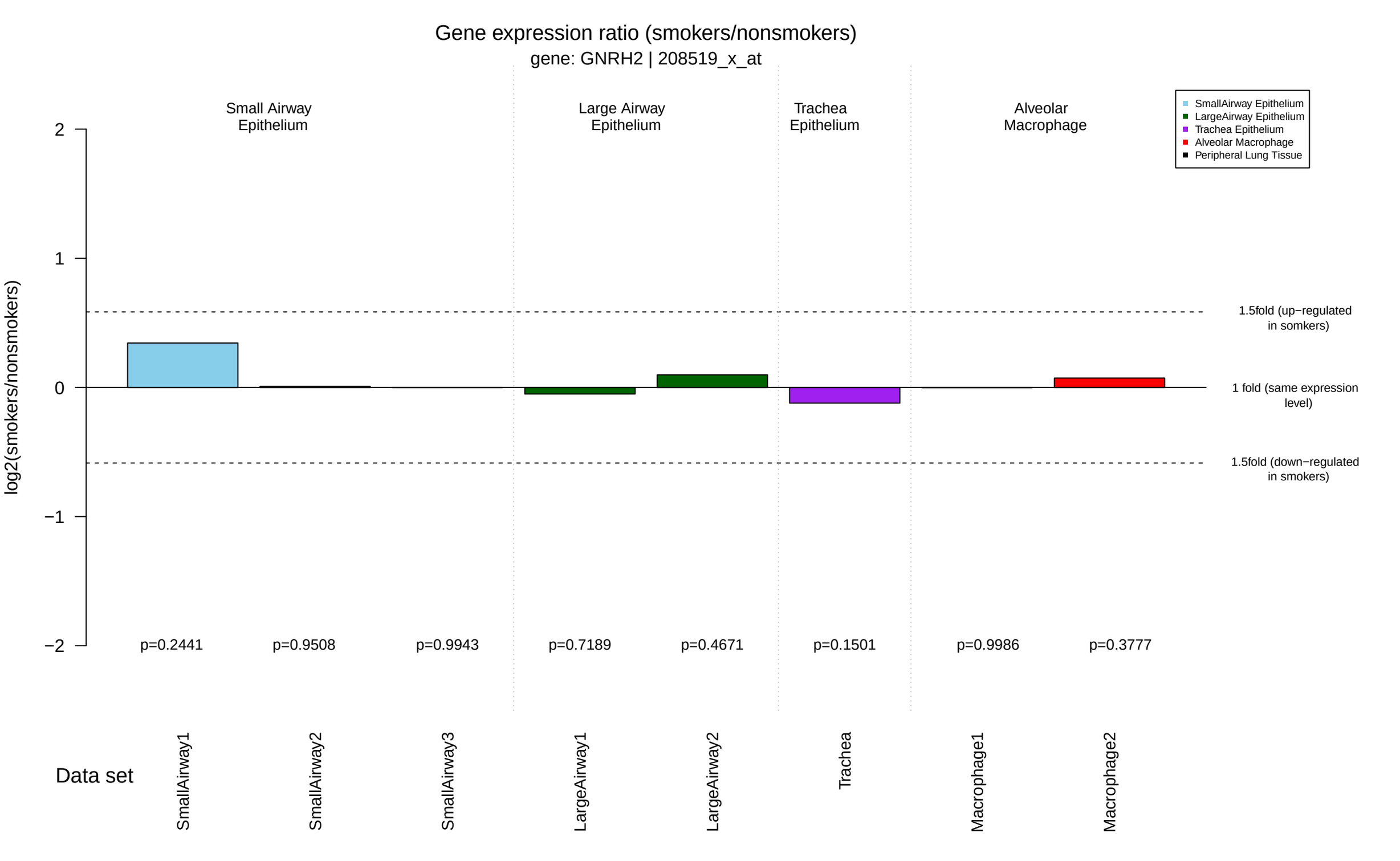
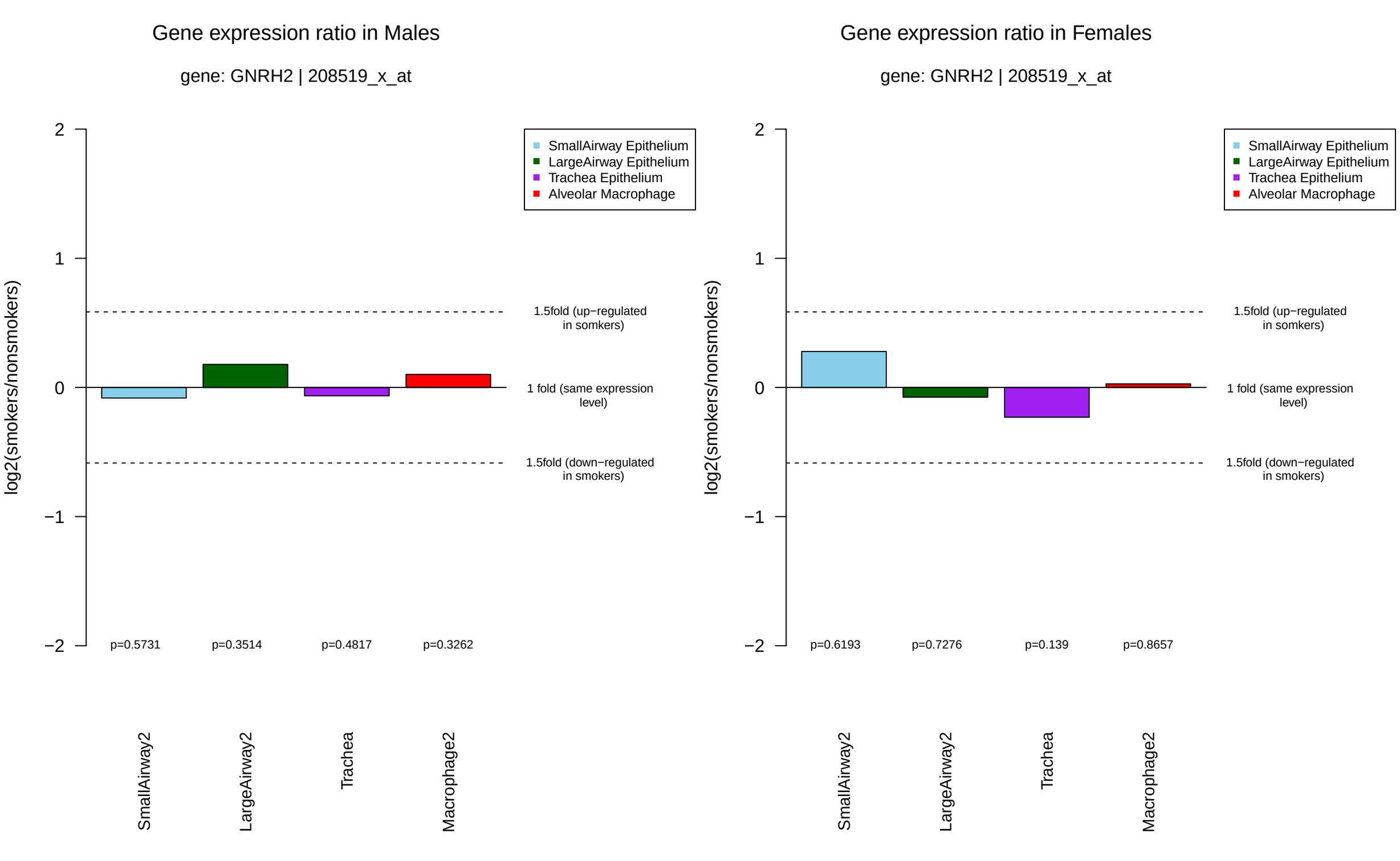
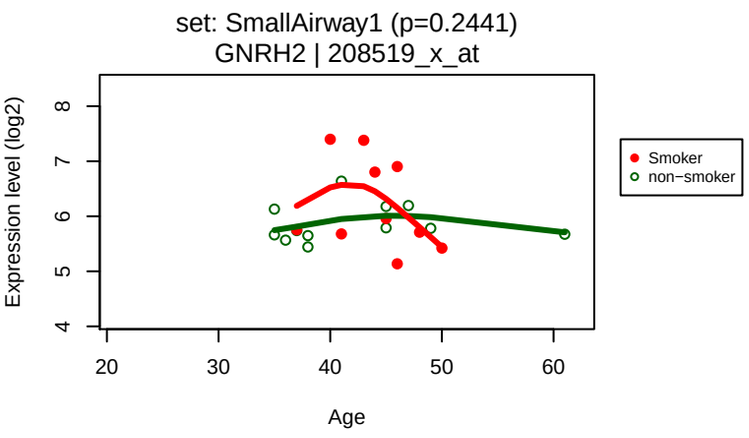
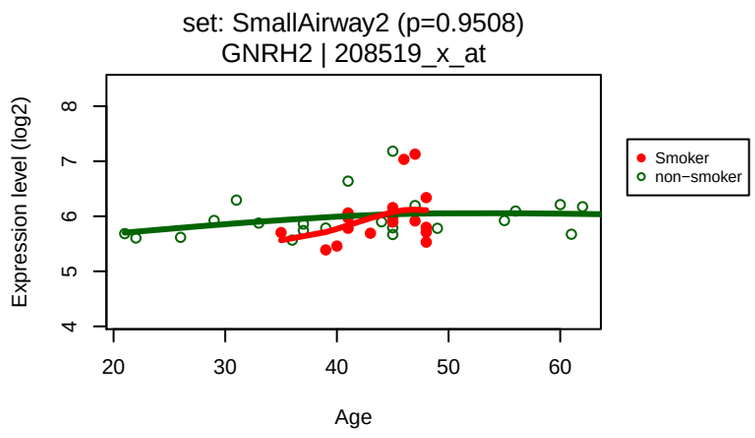
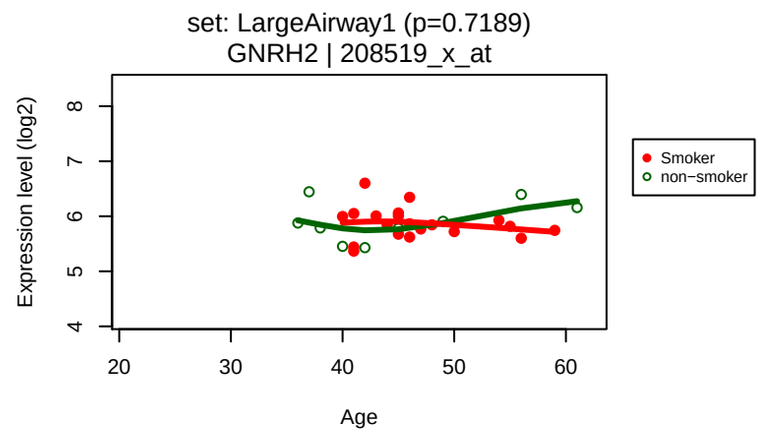

Gene expression of smokers and nonsmokers
gene: GNRH2 | 208519_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2441
p=0.9508
p=0.9943
p=0.7189
p=0.4671
p=0.1501
p=0.9986
p=0.3777
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

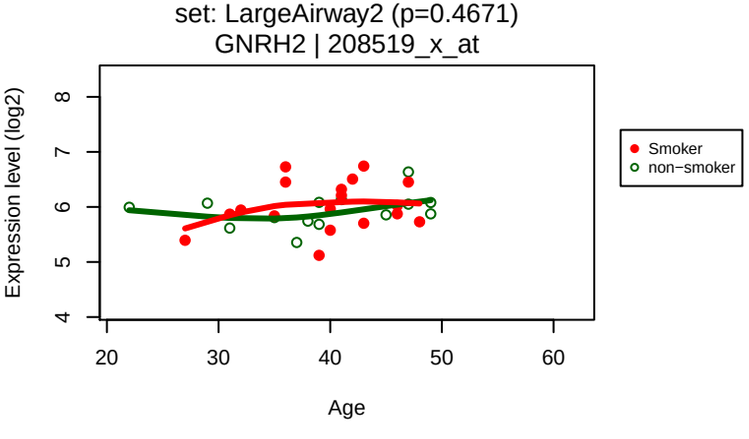

Gene expression ratio (smokers/nonsmokers)
gene: GNRH2 | 208519_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2441
p=0.9508
p=0.9943
p=0.7189
p=0.4671
p=0.1501
p=0.9986
p=0.3777
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

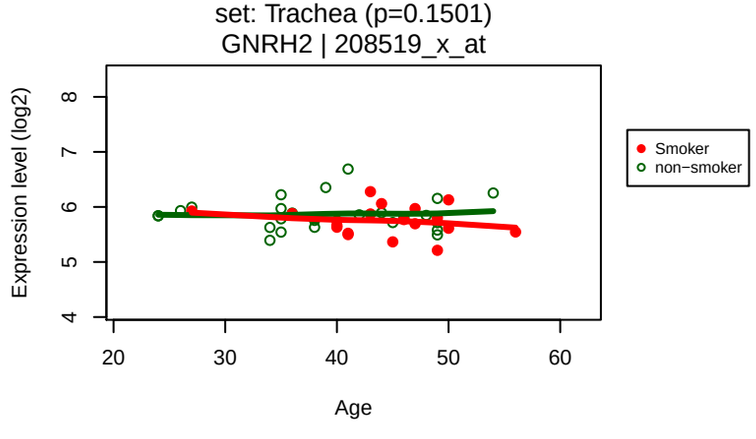

Gene expression ratio in Males
gene: GNRH2 | 208519_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5731
p=0.3514
p=0.4817
p=0.3262
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNRH2 | 208519_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6193
p=0.7276
p=0.139
p=0.8657
SmallAirway2
LargeAirway2
Trachea
Macrophage2

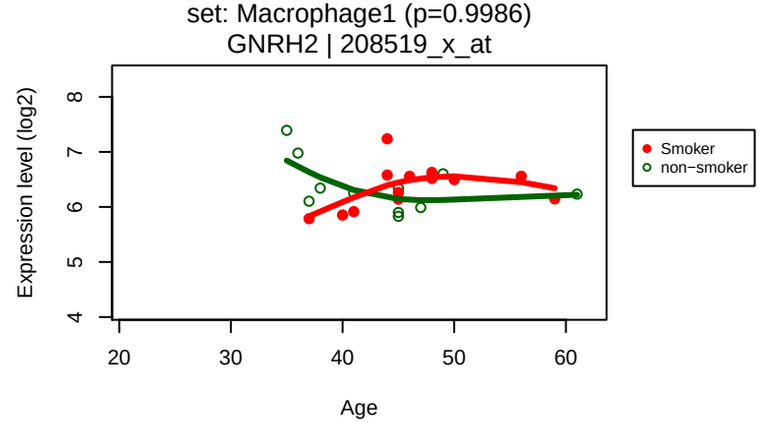

set: SmallAirway1 (p=0.2441)
GNRH2 | 208519_x_at
Expression level (log2)
Age
Smoker
non−smoker

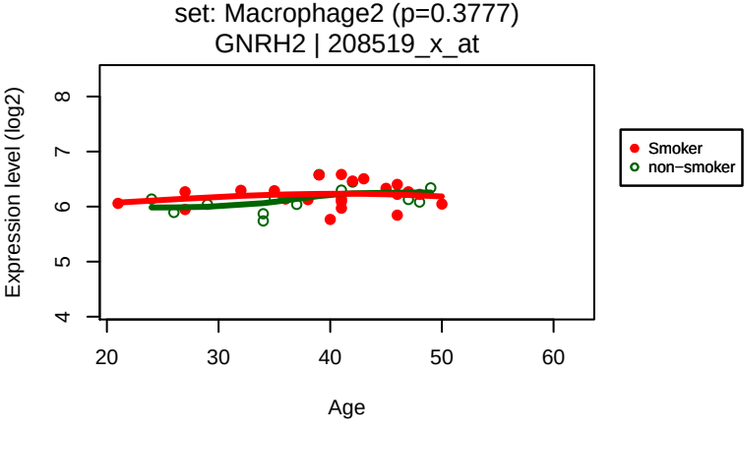

set: SmallAirway2 (p=0.9508)
GNRH2 | 208519_x_at
Expression level (log2)
Age
Smoker
non−smoker

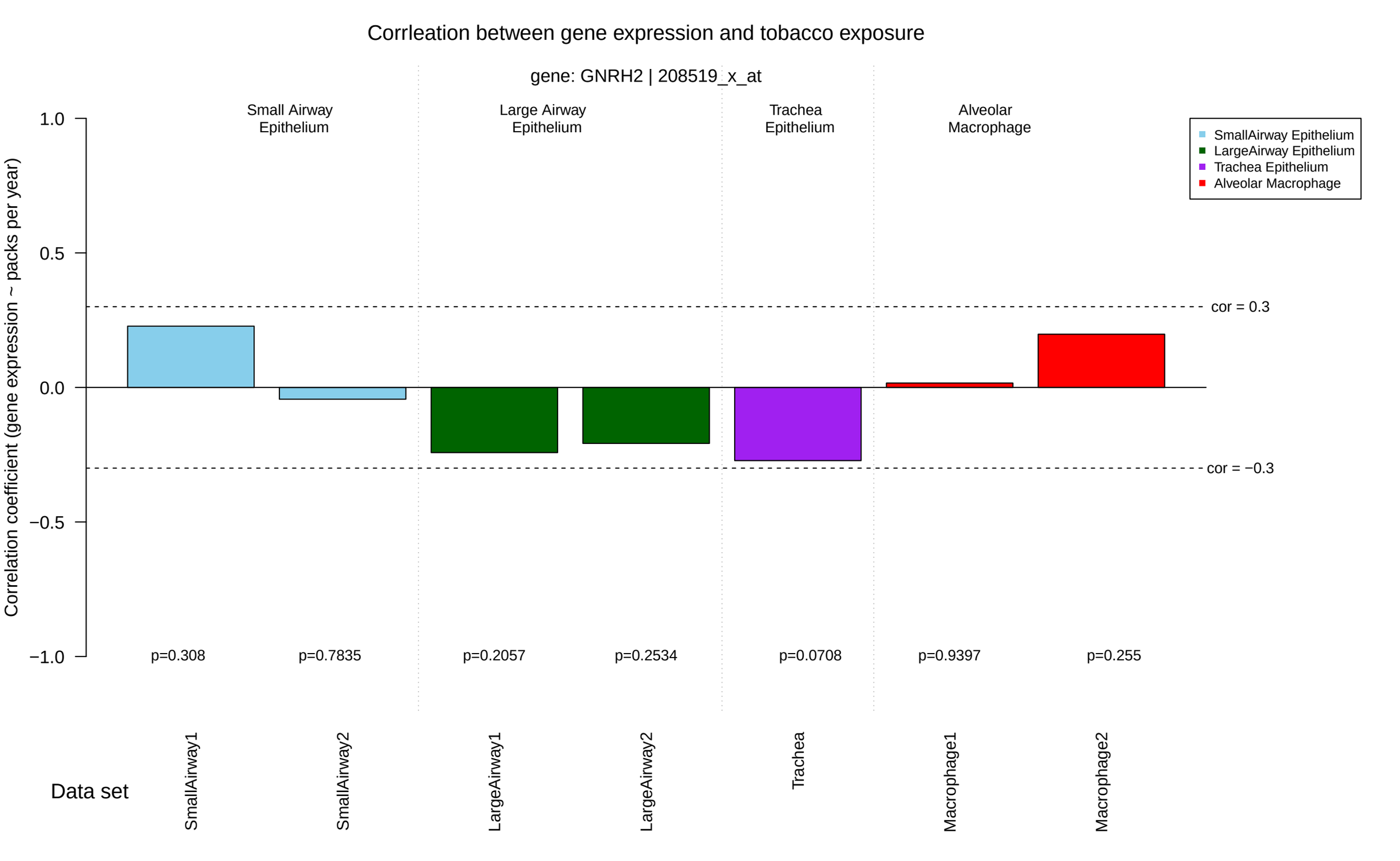

set: LargeAirway1 (p=0.7189)
GNRH2 | 208519_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4671)
GNRH2 | 208519_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1501)
GNRH2 | 208519_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9986)
GNRH2 | 208519_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3777)
GNRH2 | 208519_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNRH2 | 208519_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.308
p=0.7835
p=0.2057
p=0.2534
p=0.0708
p=0.9397
p=0.255
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2